GLOBAL FUEL MARKET ANALYSIS

Policy Choices, Price Behaviour and Economic Differences
A Tableau-Based Report | 2020–2026

REPORT OVERVIEW

This report examines the way fuel prices, government intervention and economic conditions intersect across countries. It uses a nine-view Tableau dashboard to organize the analysis into regional, operational and income-based perspectives.

Dataset scale
27,468 records | 84 countries | 7 global regions

Independent alternative report. Different structure and wording from the companion report. No images are included.

1. Abstract and Background

Fuel markets influence a wide range of economic activities. Petrol and diesel support personal mobility, freight and industrial operations, while LPG is widely used for cooking and other energy needs. When prices rise or fluctuate sharply, households may face higher living costs and businesses may experience pressure on operating budgets.

The price paid for fuel is shaped by more than the underlying commodity market. Government taxes, subsidies, regional supply conditions, currency movements and domestic policy decisions can all affect the final price. These factors make cross-country comparisons useful, but they also require careful interpretation.

Problem statement

The central challenge addressed by this project is how to interpret differences in fuel prices across countries while considering the policy and economic factors represented in the dataset. A single average cannot explain every market. Regional comparisons, time-based analysis and income-group breakdowns therefore provide complementary viewpoints.

Purpose of the study

The purpose is to use Tableau to organize a large dataset into visual summaries that help users explore regional patterns in taxation and subsidies, changes and variation in diesel prices, and differences in fuel-related indicators across income groups.

The project covers 2020 to 2026 and includes 27,468 records from 84 countries across 7 global regions. It considers petrol, diesel and LPG and contains nine visualizations designed around three stakeholder use cases.

This is an exploratory report. It describes the questions each chart can help answer; it does not claim that the charts alone establish cause and effect.

2. Dashboard Design and Analytical Structure

The dashboard moves from a broad geographical view to more focused comparisons. Its nine visualizations can be grouped into three analytical layers: policy comparison, market monitoring and economic segmentation.

Visualization catalogue

No.	Visualization	Analytical purpose
1	Global Fuel Map	View country-level geographical patterns.
2	Tax Rate by Region and Subsidy Level	Compare tax percentages across regions and subsidy groups.
3	Price vs Subsidy	Explore average fuel prices by subsidy category and region.
4	Diesel Price Forecast	Review historical movement and estimated future direction.
5	Top 10 Countries	Identify countries with the highest average diesel prices.
6	Diesel Price Volatility by Region and Year	Compare diesel-price variation over time and geography.
7	Tax by Income	Compare average tax percentages across income groups.
8	Subsidy by Income	Compare subsidy-category proportions by income group.
9	Price by Income	Compare fuel-price indicators across income groups.

How to read the dashboard

A map is useful for locating differences, while bar charts make category comparisons direct. A line chart is suited to observing movement through time. Heat maps use colour intensity to summarize many values compactly. A 100% stacked bar chart compares category composition rather than raw totals.

Dashboard filters should be used consistently. Changing the selected year, region, fuel measure or income category can alter displayed averages. Any written conclusion should identify the filter context used.

3. Policy Perspective: Taxes, Subsidies and Affordability

Stakeholder lens: Energy Policy Consultant — Dr. Sarah Mitchell

The policy perspective focuses on the relationship between public intervention and the prices observed by consumers. Governments may tax fuel to raise revenue or influence consumption. They may also subsidize fuel to reduce costs for households or selected sectors. The balance between these approaches differs across countries.

Country-level view: Global Fuel Map

The map is the entry point for geographical exploration. It helps the user see whether selected fuel indicators appear concentrated in particular parts of the world or whether neighbouring countries show contrasting values. Such differences can guide the selection of countries for more detailed comparison.

Regional comparison: Tax Rate by Region and Subsidy Level

The grouped bar chart compares average tax percentages for regions across subsidy-level categories. The consultant can ask whether tax rates tend to be higher or lower in groups with different subsidy levels and whether the relationship appears similar across regions.

Price comparison: Price vs Subsidy

The third chart compares average fuel prices by subsidy level and region. It is useful for checking whether countries assigned to higher subsidy categories also display different average prices. A visible association does not prove that the subsidy caused the price difference. Taxes, transport costs, exchange rates, production capacity and market regulation may also matter.

Questions for policy review

- Are tax patterns consistent across regions?
- Do subsidy categories show similar price patterns in different parts of the world?
- Are there countries whose results differ substantially from regional peers?
- What additional information is needed to judge whether a policy is effective and affordable?

A full policy evaluation should consider both consumer affordability and the public cost of support. The dashboard can identify patterns for investigation, while a complete evaluation requires detailed country-level policy and fiscal data.

4. Business Perspective: Diesel Costs and Supply Chain Risk

Stakeholder lens: Global Supply Chain Manager — Marcus Thompson

For logistics businesses, fuel prices affect the cost of moving products between suppliers, warehouses and customers. Diesel is especially relevant to road freight and many commercial transport operations. A manager therefore needs to understand both the general price level and the possibility of sudden changes.

Trend monitoring: Diesel Price Forecast

The forecast visualization combines historical diesel-price movement with an estimated future path. The historical portion helps identify broad direction and periods of change. The forecast can be used as a planning reference when estimating fuel budgets or reviewing procurement schedules. It should not be treated as a certain prediction: performance depends on the method, data quality and assumptions used.

Market benchmarking: Top 10 Countries

The country ranking highlights markets with the highest average diesel prices in the selected view. This provides a quick benchmark and can help managers recognize locations where fuel costs may need special attention. The ranking alone does not show total route cost, since distances, vehicle efficiency, local fees and purchasing arrangements also influence expenditure.

Risk monitoring: Diesel Price Volatility by Region and Year

The volatility heat map summarizes how diesel-price variation differs across regions and years. More intense values indicate periods or places where prices varied more strongly according to the measure used. This can help a business decide where to review budgets more frequently, maintain contingency allowances or consider flexible purchasing arrangements.

Suggested business response

Combine the dashboard with internal fleet data. Estimate fuel consumption by route, apply alternative price scenarios and identify which routes are most exposed to price increases. This turns a broad market overview into a more operational cost-management exercise.

5. Research Perspective: Income Groups and Fuel Economics

Stakeholder lens: Energy Economics Researcher — Professor Elena Rodriguez

The research perspective compares fuel-related indicators across low-, middle- and high-income country groups. Income classification provides one way to organize countries, but it should not be treated as a complete explanation of national energy policy or consumer prices.

Tax by Income

The bar chart compares average tax percentages among the three income groups. It can show whether the groups have different average tax levels in the selected data. A researcher may use the result to develop questions about fiscal capacity, energy consumption, public revenue and policy priorities. Further analysis is needed to account for differences between individual countries.

Subsidy by Income

The 100% stacked chart compares the composition of subsidy categories within each income group. Because each bar totals 100%, it emphasizes the share of each category rather than the number of records or countries. It can help assess whether the mix of low, medium and high subsidy levels differs between income groups.

Price by Income

The heat map compares average price indicators across income groups. Its colour scale helps the reader scan for relatively higher or lower values. It can be used to explore whether fuel prices vary with income classification, while remembering that taxes, subsidies, exchange rates, import dependence, production costs and regulation may also contribute.

Further research opportunities

The dashboard could be extended by comparing individual countries within each income group, examining changes before and after policy adjustments, or testing multiple variables together. Statistical methods would help determine whether observed patterns remain after accounting for other economic and market conditions.

6. Synthesis, Recommendations and Closing Remarks

What the combined views contribute

The strength of the dashboard is its combination of perspectives. The policy charts focus on taxation and subsidies; the supply chain charts focus on diesel costs, price direction and volatility; and the research charts compare patterns across income groups. Together, they provide a structured way to explore a complex global topic without relying on a single measure.

Recommendations

For public authorities: use regional comparisons to identify unusual policy patterns, then investigate the reasons using country-specific fiscal and energy information.

For businesses: treat forecast and volatility views as planning inputs, and combine them with route-specific fuel consumption, supplier quotations and scenario-based budgets.

For analysts and researchers: document chart filters and calculation definitions, test observed patterns with appropriate statistical methods, and avoid causal conclusions from descriptive comparisons alone.

Limitations

This report is based on the project description and the information visible in the dashboard reference, rather than a direct audit of the full source dataset. The detailed data dictionary, definitions of tax and subsidy categories, volatility calculation and forecast settings were not provided. The report therefore explains the analytical purpose of the charts and avoids presenting unverified numerical findings. The dashboard should be interpreted with its selected filters and measure definitions in mind.

Conclusion

Global fuel prices are shaped by interacting policy, market and economic conditions. By bringing regional comparisons, time-series analysis and income-group analysis into one Tableau dashboard, this project offers a practical framework for exploring those relationships. Its 27,468 records, 84 countries and 7 regions provide a broad basis for comparison, while the nine visualizations help translate the data into questions that policymakers, businesses and researchers can investigate further.

END OF REPORT